Your ten-category baseline

A common structure for product facts, customer feedback and future feature requests.

> **Purpose:** help the product manager trace a customer need to an existing vehicle feature, compare evidence and prepare a decision for engineering review.

ID	CATEGORY	CONCRETE SCOPE
C01	Power & Performance	Engine/motor output, transmission, drive layout, acceleration and speed.
C02	Energy & Emissions	Fuel/electricity use, emissions, battery, range and charging.
C03	Price	Vehicle price, options, packages and documented ownership costs.
C04	Safety	Brakes, airbags, stability, suspension, steering and tyre specifications.
C05	Comfort	Seat functions, climate, ride comfort, noise, audio and access convenience.
C06	Exterior	Paint, body styling, lights, glass, roof, wheel appearance and body dimensions.
C07	Software	Operating system, screens/controls, navigation, phone integration and updates.
C08	Interior Design	Upholstery, trim, dashboard/cabin layout, materials and ambient lighting.
C09	Automation	Driver assistance, cruise/lane support, parking assistance and cameras.
C10	Miscellaneous	Cargo/towing, delivery, support, sustainability and items awaiting classification.

C01-C10 identify categories; they do not rank importance. The same categories apply to F70, G60 and G70. Values, equipment availability and customer priorities remain model- and variant-specific.

THREE SCOPE LAYERS

LAYER	MEANING
Fixed / now	A versioned baseline of reviewed facts and feature definitions from the supplied sources. Missing values can remain unknown.
Variable / later	Changing customer requirements linked to baseline features; each has its own proposal and review status.
Hypothetical / later	Possible future features and comparison scenarios, with stated assumptions and synthetic-data labels.

Categories 01-05

Fields define what can be stored. A listed field does not imply that every source contains a value.

C01 Power & Performance

Include: Power (kW), torque (Nm), 0-100 km/h time (s), top speed (km/h), engine size, transmission and driven wheels.

Customer signals: Acceleration, responsiveness, gear changes and power delivery.

Boundary: Consumption and charging belong to C02. Record engine, motor and combined system ratings separately when a source distinguishes them.

C02 Energy & Emissions

Include: Fuel (L/100 km), electricity (kWh/100 km), CO2 (g/km), range (km), battery (kWh), charging power (kW) and time (min).

Customer signals: High consumption, inadequate range, slow charging or charging difficulty.

Boundary: Keep WLTP values separate from measured or reported use. Store charging start/end percentages and test conditions. Plug-in charging applies only to compatible variants.

C03 Price

Include: Base and option prices; package prices; documented service/subscription cost with its period and assumptions.

Customer signals: Affordability, value for money, expensive options and recurring charges.

Boundary: A package keeps its actual features in their categories. Price attaches to the offer; option codes and package rules are shared relationships. Missing cost data stays unknown.

C04 Safety

Include: Brake equipment, airbags, emergency call, stability systems, suspension, steering and tyre specifications. Braking distance needs a measured source and conditions.

Customer signals: Braking concerns, stability problems, airbag complaints or suspension-related safety concerns.

Boundary: Suspension is placed here as requested, with Comfort/Performance tags as needed. Driver-assistance functions are owned by C09 and can carry a Safety tag.

C05 Comfort

Include: Seat adjustment/heating/ventilation, climate zones, audio equipment, key access and tailgate convenience; cabin noise only with measurement conditions.

Customer signals: Seat discomfort, harsh ride, cabin noise, climate issues or inconvenient access.

Boundary: Seat material and visual design belong to C08. Keep reported ride/noise experience separate from published equipment and measured tests.

Categories 06-10

Fields define what can be stored. A listed field does not imply that every source contains a value.

C06 Exterior

Include: Paint/finish, exterior trim, light units, glazing, roof, wheel design and body dimensions. Preserve whether width includes mirrors.

Customer signals: Styling preferences, colour requests, lighting appearance and exterior finish concerns.

Boundary: Wheel/tyre technical specifications remain in C04. Lighting safety effects may carry a secondary Safety tag; each fact has one primary category.

C07 Software

Include: Operating-system version, display/control functions, navigation, phone connectivity, voice control and documented software-update capability.

Customer signals: Bugs, confusing menus, connection failures, app issues or update requests.

Boundary: Digital-interface functions belong here. Automated driving/parking stays in C09 even when software enables it. A reported bug does not establish its technical cause.

C08 Interior Design

Include: Upholstery/material codes, cabin trim, dashboard layout, finishes and ambient-lighting equipment.

Customer signals: Material quality, interior colour preferences, layout or finish complaints.

Boundary: Seat/climate functionality stays in C05. A seat-material issue has C08 as primary; a seating-comfort issue has C05. Both can reference the same seat feature.

C09 Automation

Include: Cruise/lane support, collision-assistance functions, parking manoeuvres, camera functions and their documented operating conditions.

Customer signals: Unreliable assistance, parking difficulty, unwanted interventions or requests for assistance features.

Boundary: Category 9 is interpreted as Automation. This covers vehicle assistance; manufacturing automation is a later implementation topic. Keep published capability wording and limits.

C10 Miscellaneous

Include: Cargo volume, weight/towing ratings, delivery services, support and sourced recycling claims. Use a named subcategory such as utility, ownership or sustainability.

Customer signals: Storage/towing needs, delivery concerns, service experiences or uncategorised requests.

Boundary: Use other_pending_review only with the original wording, review reason and owner. Repeated topics can become a new subcategory through a versioned change.

Evidence behind the baseline

Source page references are navigation aids. Every saved fact still needs an exact evidence location.

SCOPE	F70 PDF PAGES	G60 PDF PAGES	G70 PDF PAGES
C01 Power & Performance	4, 20	4-5, 21-24	4-5, 24-25
C02 Energy & Emissions	4, 20	4-5, 21-24	4-5, 24-25
C03 Price	4, 8-18	4-5, 9-19	4-5, 10-22
C04 Safety	5-7, 11, 13, 17, 20	6-7, 13-14, 18	6-8, 10-12, 21-22
C05 Comfort	5-7, 15, 17-18	6-8, 16-19	6-9, 17-22
C06 Exterior	5-6, 11, 13-14, 17, 20-21	6-7, 13-15, 18-19, 21-26	6-7, 10-12, 16, 21-22, 24-25
C07 Software	6, 17	7, 17-18	7, 21
C08 Interior Design	5, 15, 18	6, 8, 16, 18-19	6, 9, 17-22
C09 Automation	7, 17-18	7-8, 18	8-9, 22
C10 Miscellaneous	20-23	21-28	24-27

How to read this: 4, 20 means PDF pages 4 and 20; 4-5 means pages 4 through 5. Counts start at the first page of each original OptionList PDF. Ranges cover relevant sections, not proof of every proposed field.

EXCEL FEEDBACK COVERAGE

MODEL	TAGGED ROWS	DISTINCT FEEDBACK IDS	MAIN COUNTRIES
F70	4,076	2,128	GB/JP
G60	5,005	4,000	US/KR
G70	4,690	4,000	US/KR

Each file uses **Feedback_Explorer**. Repeated IDs carry multiple tags; they are not extra customer records. Distinct IDs are not verified distinct people. The files are supplied hackathon input; their real-world origin is unverified.

COVERAGE LIMITS

F70: the supplied catalogue describes petrol/diesel variants, including some mild hybrids. Its Electric Range feedback tags do not establish plug-in or electric-vehicle capability.

G70: PHEV/diesel entries occur in price pages 4-5; the detailed equipment/technical tables reviewed cover i7 and 740 xDrive. Other variant details need another source.

Context: July 2026 German-language catalogues use EUR and 19% VAT; feedback spans other countries. Match market/variant before comparison. Unspecified country applicability stays unknown.

A database that can grow

Keep category definitions stable while adding brands, sources, features, observations and decisions.

> **Shared context:** brand, model, variant, model year, powertrain, market and source edition belong to the record. Customer feedback is an evidence layer that can map to any of the ten categories.

RECORD GROUP	MINIMUM CONTRACT
Vehicle identity	Brand -> model -> variant; use internal IDs. Keep source names and model codes as aliases. Unknown variant remains explicit.
Category / feature	Stable C01-C10; named subcategories; one canonical feature/attribute ID, primary category, optional secondary tags and taxonomy version.
Source / evidence	Filename or URL, format, publisher/platform, source date, retrieval date, language, market, checksum and exact PDF page / sheet-row / web-post location.
Fact / measurement	Vehicle + field + typed value/unit + conditions + evidence + review state. Separate source-published values, measured tests and customer claims.
Feature / offer / rule	Availability by variant/market/date; brand/edition-specific option code; price amount, currency, tax basis; package contents, prerequisites and exclusions.
Feedback / mapping	Original text, source-scoped external ID, original tags, sentiment/type and country; many reviewed links from a record to features/issues.
Request / scenario	Requested feature, linked baseline, assumptions, scope layer and review decision. Store synthetic generator/version and input lineage where applicable.
Priority / decision	Issue + cohort + evidence snapshot + component scores + scoring/model version + explanation + reviewer outcome + feasibility status.

RULES THAT PRESERVE MEANING

Missing data: unknown means the evidence is silent. Not applicable requires a reason. Unavailable requires an explicit source statement.

One owner: a canonical feature has one primary category. Secondary tags capture effects such as suspension -> Comfort, while deduplication counts the record once per metric.

Versions: retain source editions and old mappings. Add subcategories/features through a reviewed taxonomy change; do not rewrite past priority runs.

New requests and comparisons

The future ingestion path preserves the difference between a specification, a report and a scenario.

WHEN A NEW SOURCE ARRIVES

STEP	RESULT
1. Capture	Store the original PDF, workbook, web item or permitted excerpt with its source ID, date and evidence location.
2. Label	Record evidence kind and data origin separately: published specification, customer statement, benchmark measurement or generated scenario; observed, synthetic or unknown origin.
3. Match	Resolve brand/model/variant when supported. Deduplicate within the source and flag likely cross-posts. Preserve original units and wording.
4. Map	Match a canonical feature under C01-C10. A new request creates a candidate feature with availability unknown; an unclear item enters C10 other_pending_review.
5. Review	Approve the meaning and mapping, or retain ambiguity. Keep contradictory reports as separate observations with their evidence.
6. Publish	Expose reviewed baseline facts and customer signals to the web app; retain unresolved items in a visible review queue.

MERCEDES / BYD SYNTHETIC COMPARISONS

Use the same feature IDs and units for each brand. Compare a defined vehicle/variant, market, time and measurement method. Keep brand-specific names as aliases, and use **not comparable** when conditions differ or evidence is missing.

Generated Mercedes/BYD rows must show **Synthetic scenario**, with the generator/model version, prompt or rule, assumptions, input references and generation date. They can test the comparison workflow; they cannot establish actual competitor specifications or customer demand.

> **Illustrative request, not supplied evidence:** 'The cabin is too noisy and I want ventilated seats' produces two issue links under Comfort: cabin noise and seat ventilation. It remains one feedback record. The ventilation request does not establish whether that feature is currently available on the customer's variant.

Future Reddit and benchmarking connectors should retain post/test URLs, source dates and collection methods. The collection method can change without changing the category or evidence structure.

Customer priority comes later

Proposed decision policy. No customer ranking or numerical score has been computed in this revision.

> **Prioritise a specific issue or feature within a model/market cohort.** A category such as Safety or Software can contain both urgent and low-priority issues. Category order is not customer priority.

PROPOSED BAND	INTERPRETATION
Critical review	Potential serious safety impact or similarly severe harm warrants prompt expert review even with few reports. The label flags review; it does not confirm a defect.
High	Substantial customer impact with credible, repeated unmet need or significant failure evidence; threshold to be calibrated with the PM.
Medium	Meaningful improvement opportunity with moderate impact or narrower demand, supported by enough evidence to assess.
Low	Limited impact or weak demand after sufficient review; retain evidence and reassess as new information arrives.
Unscored	Too little evidence, unclear mapping, incompatible cohorts or simulation-only data. Missing evidence is not automatically low priority.

INPUTS THE LATER AI MUST EXPOSE

Demand: distinct eligible feedback records mentioning the issue / all eligible feedback records in the same model, market, powertrain and time cohort. Report the numerator and denominator.

Meaning: retain Likes, Wants, Difficult to Use and Defect separately; a positive mention is not automatically an unmet requirement. Preserve both defect reports and unverified claims.

Impact and confidence: severity, evidence quality, source diversity and recency where available. Leave missing dates or severity explicit; do not invent them.

Origin: show observed-only, supplied hackathon/unknown-origin and synthetic scenario results separately. Multiple tags must not multiply one record's weight.

Decision: store the evidence snapshot, scoring/model version, component scores, explanation and PM review. Engineering cost, effort and manufacturing feasibility require separate input.

Begin with a transparent rule-based ranking that the team can inspect. Calibrate weights and band thresholds using reviewed examples before trusting AI-generated ordering. Validate feature mapping and explanations against a held-out, human-labelled sample.

From baseline to prototype

Delivery sequence for the product manager's application; technology choices can be added later.

BUILD SEQUENCE

PHASE	REVIEWABLE OUTCOME
1. Baseline	Adopt C01-C10, create shared identities/source records, extract reviewed variant facts and map workbook feedback. Display unknowns and source links.
2. New evidence	Add web/benchmark connectors and a mapping-review queue. Introduce variable requests without changing existing facts.
3. Comparison	Add brand/variant matches and clearly labelled Mercedes/BYD synthetic scenarios. Compare only compatible fields and conditions.
4. Prioritisation	Implement versioned rules and then AI-assisted mapping/ranking. Review explanations, measure quality and retain human decisions.
5. Web app / agents	Give the PM filters, evidence drill-down, comparisons, priority rationale and approvals. Different models can share the same reviewed data and tool interfaces.

Planned integration: Web application -> orchestration -> ingestion, retrieval, comparison and priority tools -> canonical database. An MCP interface can be considered at the tool boundary. Record the model/tool versions behind each recommendation.

BASELINE ACCEPTANCE CHECKS

1. Exactly ten primary categories; every legacy field has a defined destination in a category, shared metadata, feature relationships or the feedback layer.

2. Every displayed product value resolves to the right variant/market/edition and an exact source location; missing values remain explicit.

3. A repeated Excel ID creates one source-scoped feedback record with multiple tags; unmapped records remain visible.

4. Synthetic and unknown-origin records retain their labels through imports, comparisons, exports and priority views.

5. Priority results have an evidence snapshot, cohort, denominator and version. Changes in source data do not silently rewrite an earlier decision.

SOURCE REGISTER AND DELIVERY STATUS

Primary files: F70_OptionList.PDF, G60_OptionList.PDF, G70_OptionList.PDF and the three corresponding *_feedback_hackathon.xlsx workbooks. July 2026 editions: F70 issue 1 (28 May), G60 issue 2 (22 June), G70 issue 1 (28 May). The teammate's BMW_G60_Scopes_and_Data.pdf remains a derived G60 reference.

Delivered: revised scope PDF, matching v2 taxonomy/field map and ten-category seeds in the existing BMW starter schema. The database is not populated. Multi-brand/web/synthetic-lineage tables, scraping, comparison data, AI scores and the application are future implementation work; the starter SQL is not a migration for an existing database.